

LAM PHAM

SENIOR SOFTWARE ENGINEER

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Work Experience

NETKO, AI ENGINEER

Ha Noi, Mar. 2025 - Feb.2026

- Architected and built the backend for a scalable **microservices-based AI multi-agent system** enabling intelligent automation for **supply chain optimization** and **IT workflows**. Delivered high availability and low-latency inference; the product was showcased at the *Supply Chain Design Conference Tokyo 2025*.
- Designed hybrid SQL and vector database schemas with strict **tenant isolation controls** to secure multi-tenant environments. Optimized RAG-based memory retrieval to achieve **sub-300ms P95 vector search latency** and **sub-2s P99 end-to-end RAG latency** with zero cross-tenant data leakage.
- Implemented a **distributed asynchronous task queue** with non-blocking async I/O (`httpx`) to manage **batched LLM inference and I/O-bound workloads**. Effectively handled burst traffic without dropping requests, achieving a throughput of **1,000+ inferences/min** and **sub-20ms API task dispatch latency**.
- Engineered a high-performance inference stack using **vLLM and Text Embeddings Inference (TEI)**, integrating queued task processing for high-concurrency workloads. Achieved **sub-300ms P99 Time-To-First-Token (TTFT)**, **50+ Tokens Per Second (TPS)** per request, and an overall throughput of **250+ TPS**, significantly optimizing GPU compute utilization.
- Developed an end-to-end observability framework using **Grafana, Prometheus, and OpenTelemetry (Jaeger)** to monitor distributed traces, agent state transitions, and high-cardinality metrics. Leveraged trace complexity-workflow to identify LLM bottlenecks and resolved LLM bottlenecks using trace data, cutting inference latency by **75%** and reducing overall agent wait times by **33%**.
- **Tech Stack:** FastAPI, PostgreSQL, Milvus, Redis, LlamaIndex, LangGraph, Celery, RabbitMQ, Docker, vLLM, Grafana, Prometheus, OpenTelemetry, MinIO, Locust.

Education

Hanoi University of Science and Technology

Hanoi, Vietnam

B.E. IN ELECTRICAL, ELECTRONICS AND COMMUNICATIONS ENGINEERING (GPA: 2.9/4.0)

Sep. 2014 – Sep. 2019

Honors & Awards

- **3rd Prize** in the Company (SEA Group) Hackathon. 2020
- **Runner-up paper award** on the topic of Semantic Text Alignment based on Topic Modeling. 2016

Skills

- **Programming Languages:** Python, Java.
- **Databases:** PostgreSQL, Milvus, Redis.
- **Distributed Systems & Frameworks:** RabbitMQ, Celery, FastAPI, ORM, LlamaIndex, LangGraph.
- **DevOps & Monitoring:** Docker, vLLM, Linux, Git, Prometheus, Grafana, Jaeger.